ADP i DAP Auditorne vježbe

1.

DC napon → $U = 6\,V$

$t_{int} = 240\,\mu s$

$U_{komp} = 3V$

$t_{komp} = 50\,ms$

a) vremenski dijagram 1 ciklusa pretvorbe

b) Odredite pokazivanje brojila u slučaju zanemar. trajanja izbijanja kond

c) Koliko je maksimalno trajanje izbijanja kond. uz uvjet da je pokazivanje brojila isto kao u slučaju b)?

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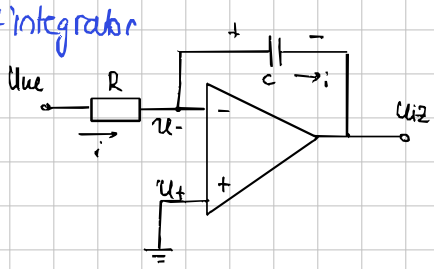
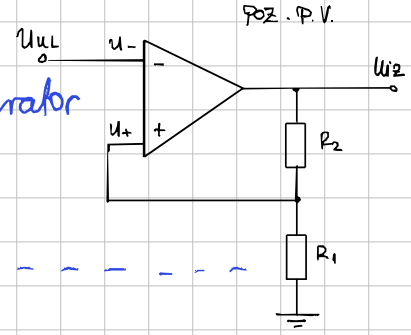
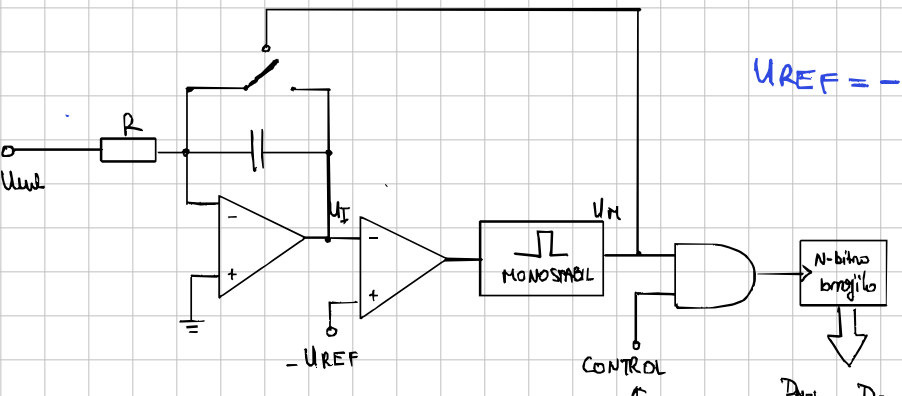
* integrator



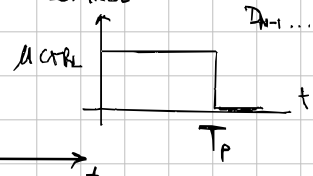
* komparator

POZ. P. V.

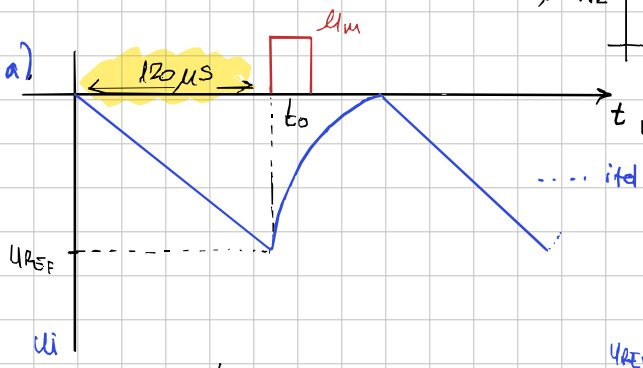


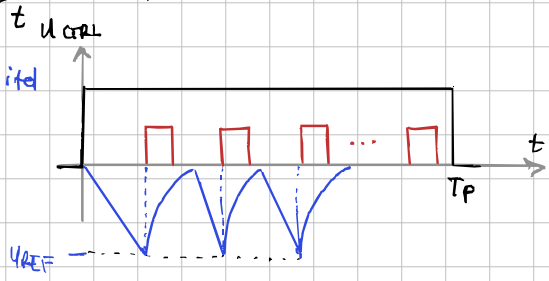


$U_{REF} = -3V$



a)





b) $u_i = -\frac{1}{RC}\int_0^{t_0} u_{ul}(t)\,dt = -\frac{1}{RC}\,U_{ul}\cdot t_0 = U_{REF}$

$t_0 = -J\cdot\frac{U_{REF}}{U_{ul}} = 240\,\mu s\,\frac{3V}{6V} \longrightarrow t_0 = 120\,\mu s \Longrightarrow f_0 = \frac{1}{t_0} = -\frac{1}{J}\,\frac{U_{ul}}{U_{REF}}$

jer je U_{REF} na minus strani

$f_0 = 8333.3\,Hz$

N = koliko ih ima u intervalu ⇒ $N = \frac{T_p}{t_0} = T_p\cdot f_0$

$N = 50\,ms\cdot\frac{6V}{240\,\mu s\cdot 3V} \Rightarrow N = 416.67 \Rightarrow 416.$

zaokruži na niže

c)

Tp sigurno traje bar N·t0 uz još neko vrijeme

$T_p = N\cdot t_0 + (N-1)\cdot t_{izb}$

$t_{izb} = \frac{T_p - N\cdot t_0}{N-1} = \frac{50\,ms - 416\cdot 120\,\mu s}{415} \longrightarrow t_{izb} = 0.193\,\mu s$

Zadatak 2.) ADP, dvostruko pilasti napon

$f_{osc} = 400\,kHz$

$U_{REF} = -10V$

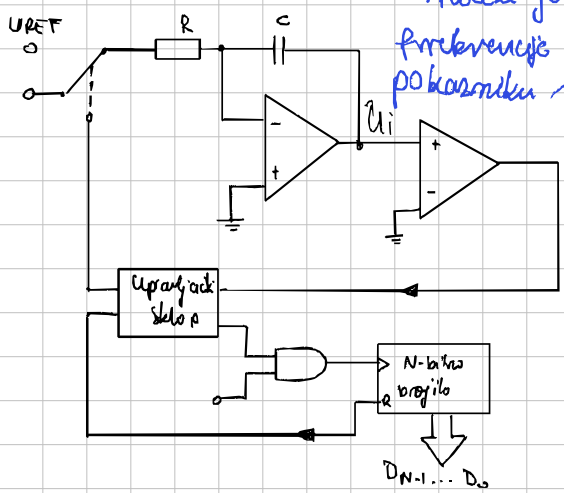
$N = 10\,000$ imp

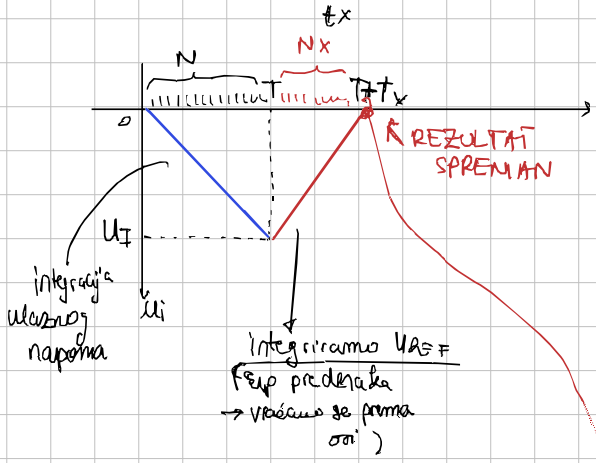
a) Vremenski dijagram jednog ciklusa pretvorbe

b) Najduže trajanje mjernog ciklusa

c) Ulazni napon pri kojem ↕ vrijedi

d) Koliki je U_{ul} ako se uz superponiranu smetnju frekvencije 50Hz i amplitude 500mV na pokazniku spojenom na izlaz pretvornika očita 3,755V nakon 1 ciklusa pretvorbe.





$T = \frac{N}{f_{osc}}$ $\quad t_x = \frac{N_x}{f_{osc}}$

=> šalje se signal upravljačkom sklopu da je gotov ciklus

$$u_i(T) = U_I = -\frac{1}{RC}\int_0^T u_{ul}(t)\,dt$$

* odmaži

↓

može biti DC, DC sa superponiranom smetnjom ili izmjeničan

$$u_i(T) = \frac{-1}{RC}\,T\cdot U_{UL} \rightarrow \frac{1}{T}\int_0^T u_{ul}(t)\,dt$$; srednja vrijednost ulaznog signala u intervalu T

→ integrator nastavlja integrirati ulazni napon do trenutka $T + t_x$

$$u_i(T+t_x) = U_I - \frac{1}{RC}\int_T^{T+t_x} U_{ref}\,dt = U_I - \frac{t_x}{RC}U_{REF} = 0$$

// ima memoriju!

Ciklus pretvorbe će završiti kada napon integratora dosegne nulu

$$\Rightarrow \frac{-1}{RC}\cdot T\cdot U_{UL} = \frac{t_x}{RC}\cdot U_{REF}$$

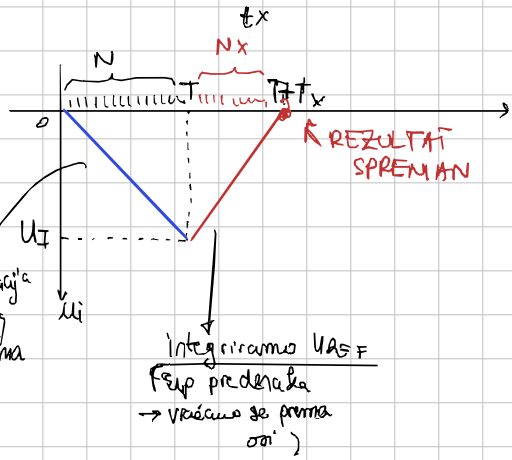
$$\boxed{t_x = T\cdot\frac{U_{UL}}{U_{REF}} \qquad N_x = N\cdot\frac{U_{UL}}{U_{REF}}}$$

$T_{CIKLUS} = T + t_x$

$t_{x,max} = T$ za $U_{UL,max} = U_{REF}$

$T_{CIKLUS,max} = 2T$

d) Koliki je U_{UL} ako se uz superponiranu smetnju frekvencije 50Hz i amplitude 500mV na okosniku spojenom na izlaz prekvomika očita 3,755V nako 1 ciklusa pretvorbe.



$$U_{UL} = \frac{1}{T}\int_0^T u_{ul}(t)\,dt$$

$$= \frac{1}{T}\int_0^T \Big(\underbrace{U_{DC}}_{\text{istosmjer.}} + \underbrace{U_0 \cdot \sin(2\pi f \cdot t + \alpha)}_{\text{izmjenična}}\Big)dt$$

$$U_{UL} = U_{DC} + \frac{U_0}{2\pi f t}\cos(2\pi f \cdot t + \alpha)\Big|_0^T$$

$$U_{UL} = U_{DC} - \frac{U_0}{2\pi f t}\left[\cos(2\pi f T + \alpha) - \cos(\alpha)\right]$$

$$U_{UL} = U_{DC} + \frac{U_0}{\pi f t}\cdot\sin(\pi f T + \alpha)\cdot\sin(\pi f T)$$

=> šalje se signal upravljačkom sklopu da je gotov ciklus

$U_{UL} = 3{,}755\,V$ → zadano, ali ne znamo ništa drugo

$$\pi f T = \pi \cdot 50\,Hz \cdot \frac{10000}{400 \times 10^3}$$

$$\rightarrow \pi f T = \frac{5\pi}{4} \overset{*}{\approx} (\text{sinus}) \Rightarrow -\frac{\sqrt{2}}{2}$$

$$U_{DC} = U_{UL} \pm \frac{U_0}{\pi f T}\cdot\frac{\sqrt{2}}{2}$$

$$U_{DC} = 3.755\,V \pm \frac{500\,mV}{\pi \cdot 50\,Hz \cdot \frac{10000}{400\,kHz}}\cdot\frac{\sqrt{2}}{2}$$

$U_{DC} = 3.755\,V \pm 90{,}0\,mV$ ovisno o početku vremena pretvorbe i kraju pretvorbe

Zadatak 4.) ADP

dva pilasta napona, $f_{os} = 1\,MHz$, $U_{REF} = 5V$, $N = 10\,000$ imp

$N_2 = ?$ $U_{UL} = 3.6V$

$N_3 = ?$ (u slučaj da nema smetnje)

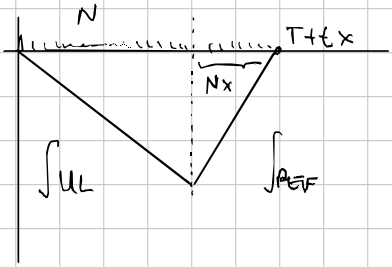
smetnja: Trokutasti valni oblik
$U_S = 180\,mV$
$f_S = 125\,Hz$
(početak pretvorbe sink. sa smetnjom)

bez smetnji: $U_{UL} = 3.6V$

$N = 10^4$

$$N_x = -N \cdot \frac{U_{UL}}{U_{REF}} = -10^4 \cdot \frac{3{,}6}{5} \rightarrow N_x = 7200$$

$U_{REF} = -5V$



$$T = \frac{1}{f_{osc}} \cdot N = \dots = 10\,ms$$

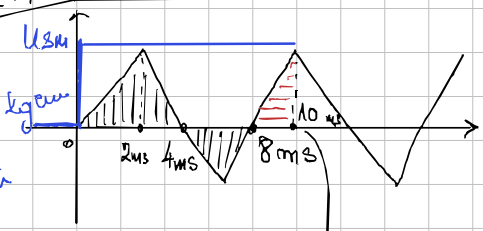
Sa smetnjom → sredujemo vr. signala u intervalu T

$$N_x = -N \cdot \frac{\overline{U_{UL}}}{U_{REF}}$$

trebamo sr. vr. $\overline{U_{UL}} = \frac{1}{T}\int_0^T u_{UL}(t) = U_{UL} + \frac{1}{T} \cdot \frac{\Delta T \cdot U_{sm}}{2}$

$u_{UL}(t) = U_{UL} + u_{sm}(t)$

smetnja je trokutastog oblika



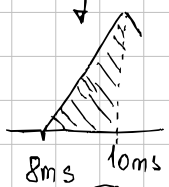
vrijeme u kojem integriramo smetnju

$f_{sm} = 125\,Hz$

$\rightarrow T_{sm} = 8\,ms$

$$N_x = -N \cdot \frac{\overline{U_{UL}}}{U_{REF}}$$

vrijeme u kojem integriramo smetnju!



$\underbrace{8ms \quad 10ms}_{2ms = \Delta S}$

$$\Rightarrow \overline{U_{UL}} = 3{,}6V + \frac{1}{10ms} \cdot \frac{2ms \cdot 180mV}{2}$$

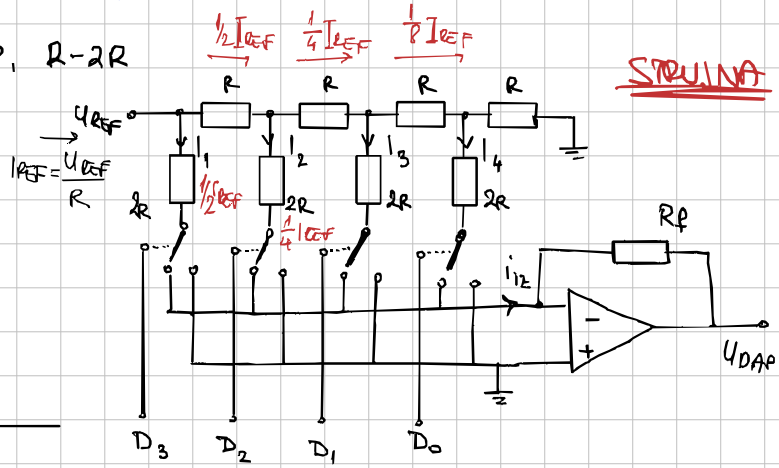
$\overline{U_{UL}} = 3{,}618V \Rightarrow N_x' = -10000 \cdot \frac{3{,}618}{-5} \Rightarrow \boxed{N_x' = 7236}$

2. dio AV

DAP; naglasak na R-2R

Zadatak 2.1) DAP, R-2R

STRUJNA



$R = R_f = 15\,k\Omega$

$U_{REF} = -5V$

$D_3 D_2 D_1 D_0 = 1101$

$u_{iz} = ?$

$U_{DAP} = -R_f \cdot i_{iz}$

$$U_{DAP} = \frac{-U_{REF}}{2^N}\left(D_{N-1}\cdot 2^{N-1} + D_{N-2}\cdot 2^{N-2} + \dots + D_0\right), \text{ za } R_f = R$$

$N = 4$ (jer su 4-bita)

$$U_{DAP} = \frac{-U_{REF}}{2^4}\left(D_3\cdot 2^3 + D_2\cdot 2^2 + D_1\cdot 2 + D_0\right) = \frac{-U_{REF}}{16}\left(1\cdot 8 + 1\cdot 4 + 0 + 1\right)$$

$$U_{DAP} = -\frac{13}{16}\cdot U_{REF} = \frac{-13}{16}\cdot 5V \longrightarrow \boxed{U_{DAP} = -4.0625V} = u_{iz}$$

Zadatak 2.2.)

DAP, R-2R

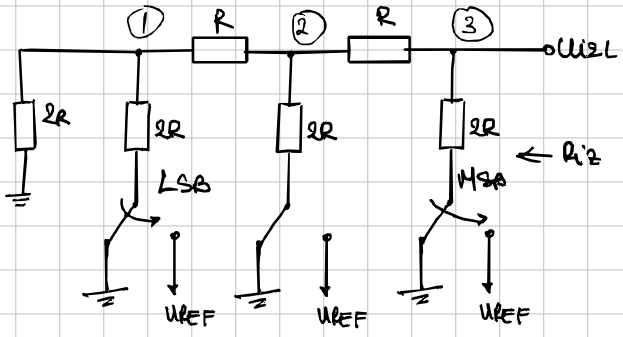
NAPONSKA

$U_{REF} = 5V$

Kvant korak?

$R_{iz} = ?$

Strujno opterećenje



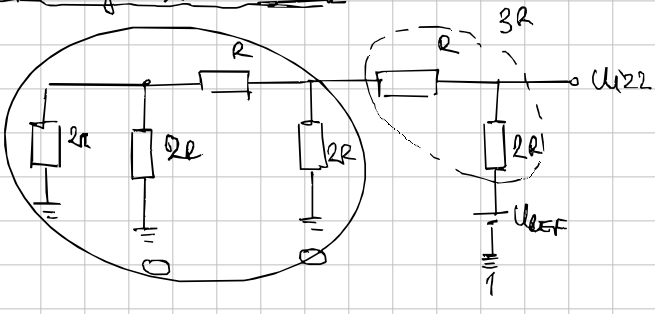
vidimo da je $R_{iz} = R = 250\,\Omega$

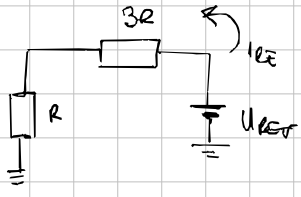
+mana → strujno opt ovisi koji smo dig. podatak odabrali

U_{iz} → fiksna struja ⇒ ne ovisi o digit. podatku

Opterećenje za 100_2

MSB LSB

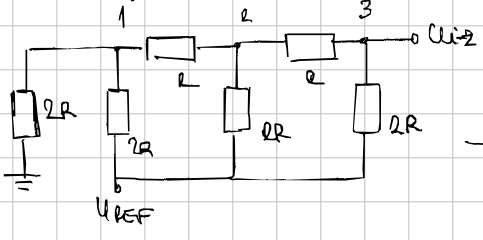




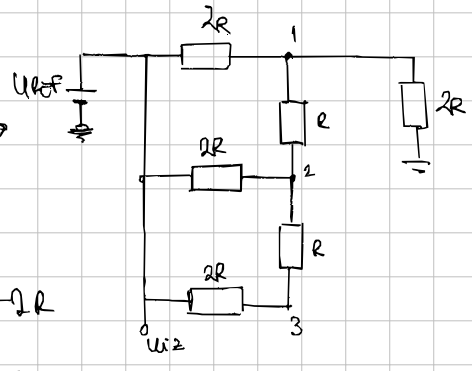
$$I_{REF}(100_2) = \frac{U_{REF}}{4R} = \frac{5V}{1000\,\Omega}$$

$$\boxed{I_{REF}(100_2) = 5mA}$$

Opterećenje za 111_2



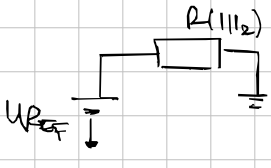
→



$$R(111_2) = \left[(3R \parallel 2R) + R\right] \parallel 2R + 2R$$

$$= \left[\frac{6R^2}{3R+2R} + R\right] \parallel 2R + 2R = \left[\frac{6}{5}R + R\right] \parallel 2R + 2R$$

$$= \frac{\frac{22}{5}R}{\frac{21}{5}} + 2R \Rightarrow R(111_2) = \frac{64}{21}R \longrightarrow R(111_2) = 761.905\,\Omega$$



→ $\boxed{I_{REF}(111_2) = 6.56mA}$

Zašto je ovo problem? → Napon reference će se mijenjati u ovisnosti koliko struju iz reference vučemo.

Koef strujne regulacije: $\Delta U_{REF} = k_i \cdot \Delta I_{REF}$

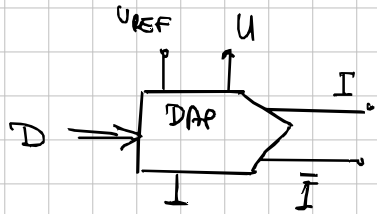
$$U_{DAP} = -\cancel{\frac{U_{REF}}{2^N}} \cdot D \longrightarrow U_{DAP} = -\frac{U_{REF}(D)}{2^N} \cdot D$$

↳ nije niže linearno

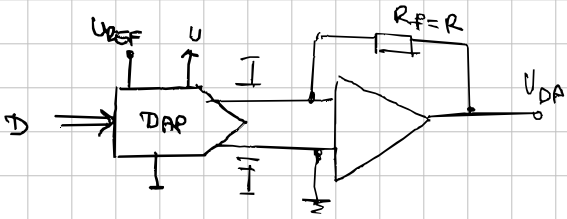
Zadatak 23.) R-2R

STRUJNI IZLAZ!

$U_{REF} = 5V$



Izvesti izlazni stupanj s 2 OP na način da opseg u_{iz} bude ±5V i to uz upotrebu bin. koda s pomakom.



$$U_{DA} = -R \cdot I$$

$$U_{DA} = -R \cdot \frac{U_{REF}}{R} \cdot \frac{1}{2^N} \underbrace{\left(d_{N-1} \cdot 2^{N-1} + \dots + d_0\right)}_{D}$$

$$U_{DA} = -\frac{U_{REF}}{2^N} \cdot D$$

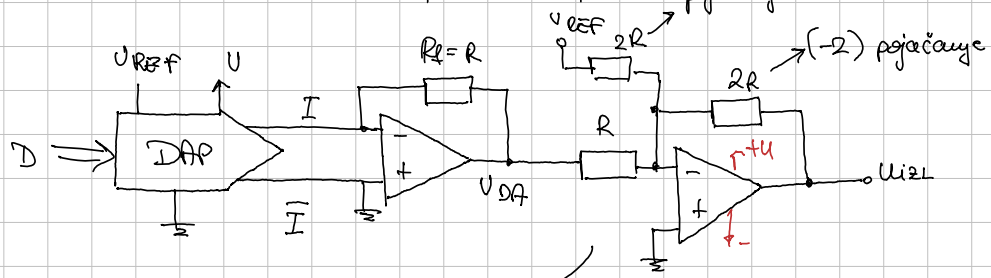
Neka je DAP 4-bitni

	D	U_{DA}/U_{REF}	$\times(-2)$	$-1\ (u_{iz}/U_{REF})$
0000	0	0	0	0
0001	1	$-\frac{1}{16}$	$+\frac{1}{8}$	$-\frac{7}{8}$
...			...	
1000	8	$-\frac{8}{16}$	$+1$	0
1001	9	$-\frac{9}{16}$	$+\frac{9}{8}$	$\frac{1}{8}$
...			...	...
1110	14	$-\frac{14}{16}$	$+\frac{14}{8}$	$\frac{6}{8}$
1111	15	$-\frac{15}{16}$	$+\frac{15}{8}$	$\frac{7}{8}$

$$u_{izL} = (-2)U_{DA} - 1$$

$$u_{izL} \Rightarrow \left[-U_{REF}, \frac{7}{8}U_{REF}\right]$$

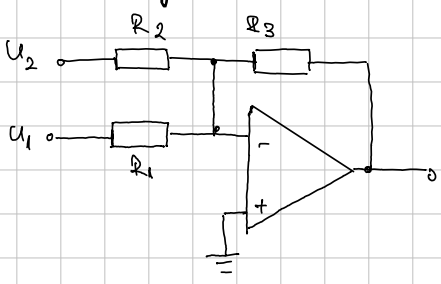
sa drugim OP treba dovršiti da je opseg do U_{REF} (ne samo $\frac{7}{8}U_{REF}$)



$$u_{iz} = \left(-\frac{2R}{R}\right) \cdot U_{DA} + \left(-\frac{2R}{2R}\right) \cdot U_{REF} = \dots \Rightarrow u_{izL} = \frac{U_{REF}}{2^{N-1}} \cdot D - U_{REF}$$

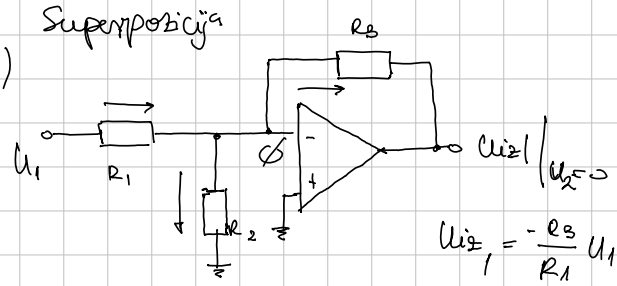
$$u_{izL} = U_{REF}\left(\frac{D}{2^{N-1}} - 1\right)$$

Podijeli se:



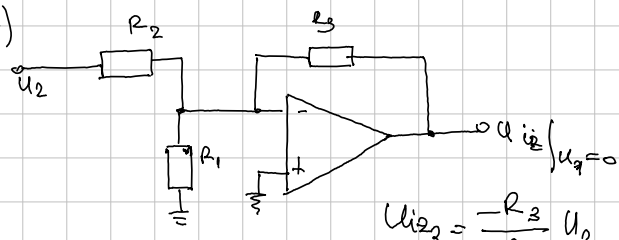
Superpozicija

1)



$$U_{iz_1} = \frac{-R_3}{R_1} U_1$$

2)

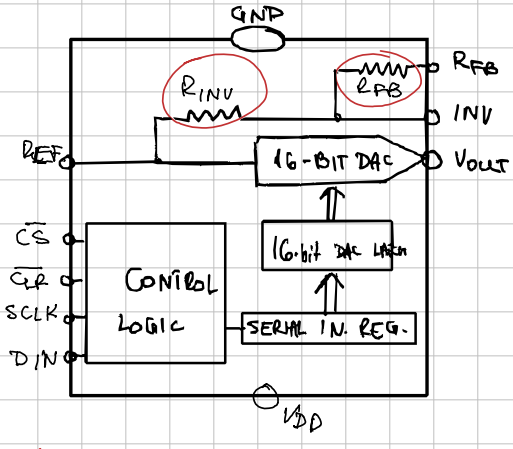


$$U_{iz_2} = \frac{-R_3}{R_2} U_2$$

$\Longrightarrow$ Sveukupno:

$$U_{iz} = \frac{-R_3}{R_1} \cdot U_1 - \frac{R_3}{R_2} \cdot U_2$$

Zadatak 2.4.) Pomoću 1 OP izvedite izlaz DAP-a AD 5542A u bipolarnom kodu s pomakom. Blok shema pretvornika dana je shemom:



U_{iz} u ovakvoj konfiguraciji:

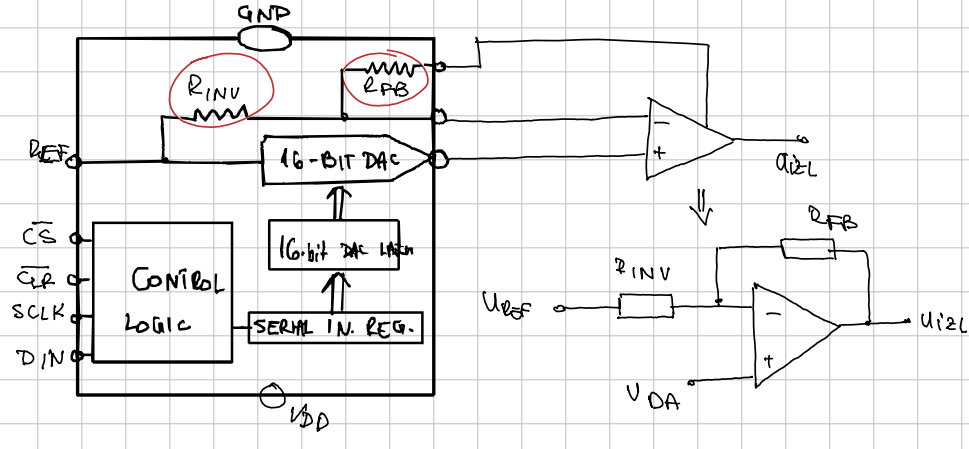
$$U_{DA} = \frac{U_{REF}}{2^N} \cdot D$$

$\downarrow$

opseg $U_{DA} = \left[0;\ U_{REF} \cdot \frac{2^{16}-1}{2^{16}}\right]$

$$U_{izL} = 2 \cdot U_{DA} - U_{REF}$$

bilo da su $R_{FB} = R_{INV}$



$$U_{izL} = -\frac{R_{FB}}{R_{INV}} \cdot U_{REF} + U_{DA}\left(1 + \frac{R_{FB}}{R_{INV}}\right), \quad R_{FB} = R_{INV}$$

$U_{izL} = 2U_{DA} - U_{REF}$ to je ono što smo trebali *

* na specifikacijama: $\pm 0.0015 \approx \frac{1}{2^{16}}$

D	U_{DA}	U_{izL}
00…00	0	$-U_{REF}$
⋮		
10…00	$U_{REF} \cdot \frac{2^{15}}{2^{16}}$	0V
⋮		
11…11	$U_{REF} \cdot \frac{2^{16}-1}{2^{16}}$	$U_{REF} \cdot \frac{2^{15}-1}{2^{15}}$

bipolarni kod s pomakom!